$$\langle F(x)\rangle \neq F(\langle x\rangle).$$

The right-hand side is a function of the center of the wave packet. It is not the same as the left-hand side, which corresponds to our results from the previous section—<$F(x)$> has the same form as the right-hand side of Eq. 9.18.[5]

Let me give you an example where these two expressions could be extremely different. Suppose that F is equal to x squared:

$$F = x^2.$$

And suppose the wave packet looks like Fig. 9.3. What's the expectation value of x? It's zero, and so is $F(\langle x\rangle)$, because $F(0) = 0^2 = 0$. On the other hand, what is the expectation value of x^2? It's greater than zero. So when a wave packet is not a nice, single bump that is mainly characterized by its center, it's not always true that the time rate of change of the momentum is the force evaluated at the expectation value of x. It's only when the wave function is concentrated over a fairly narrow range that the expectation value of $F(x)$ is the same as $F(\langle x\rangle)$. So we have cheated a little in saying our quantum equation of motion looks classical. That depends on the wave packet being coherent and well localized.

Everything else being equal, when the mass of a particle is large, the wave function tends to be very well concentrated. If there are no very sharp spikes in the potential function $V(x)$, then it will be a good approximation to replace <$F(x)$> with $F(\langle x\rangle)$. When $V(x)$ has spikes, however, the wave packet tends to break up. For example, suppose we have a nice wave packet moving to the right, and it hits a point structure, like an atom, with a potential function similar to Fig. 9.4. The wave packet will spread out and disintegrate. If, on the other hand, it hits a very smooth potential, then it will go through the smooth potential, moving more or less according to the classical equations of motion. We don't expect quantum mechanics to reproduce classical mechanics in every possible circumstance. We expect it